21SHARES · CAPITAL MARKETS & TRADING · TAKE-HOME PROJECT

Last-10-second momentum in BTC/USDT

A trading strategy built from 121 million one-second klines, January 2023 to October 2026

Siyun (Irene) Wang · November 2026

Summary

121m

one-second klines

Every 1-second BTC/USDT kline from Jan 2023 to Oct 2026, 46 monthly files, all checksums verified

0.19 bp

edge per minute, out-of-sample

2025 to Oct 2026, before costs (Newey-West t = 30). Breakeven cost about 0.2 bp per unit traded

−1.7%

worst day after risk management

Volatility-targeted sizing and a 1.5% daily loss limit, down from −7.1% for the unmanaged strategy

Idea. Since Binance ended zero-fee BTC/USDT trading on 22 March 2023, short-term price moves tend to continue for a few seconds. I trade in the direction of the **last 10 seconds** of each minute and hold for one minute; any window from 2 to 30 seconds works, and in-sample data selects 10 seconds. The edge is real and stable but smaller than exchange fees, so its most practical use is **execution timing**.

Data: 121 million rows, checked and reduced to minute bars

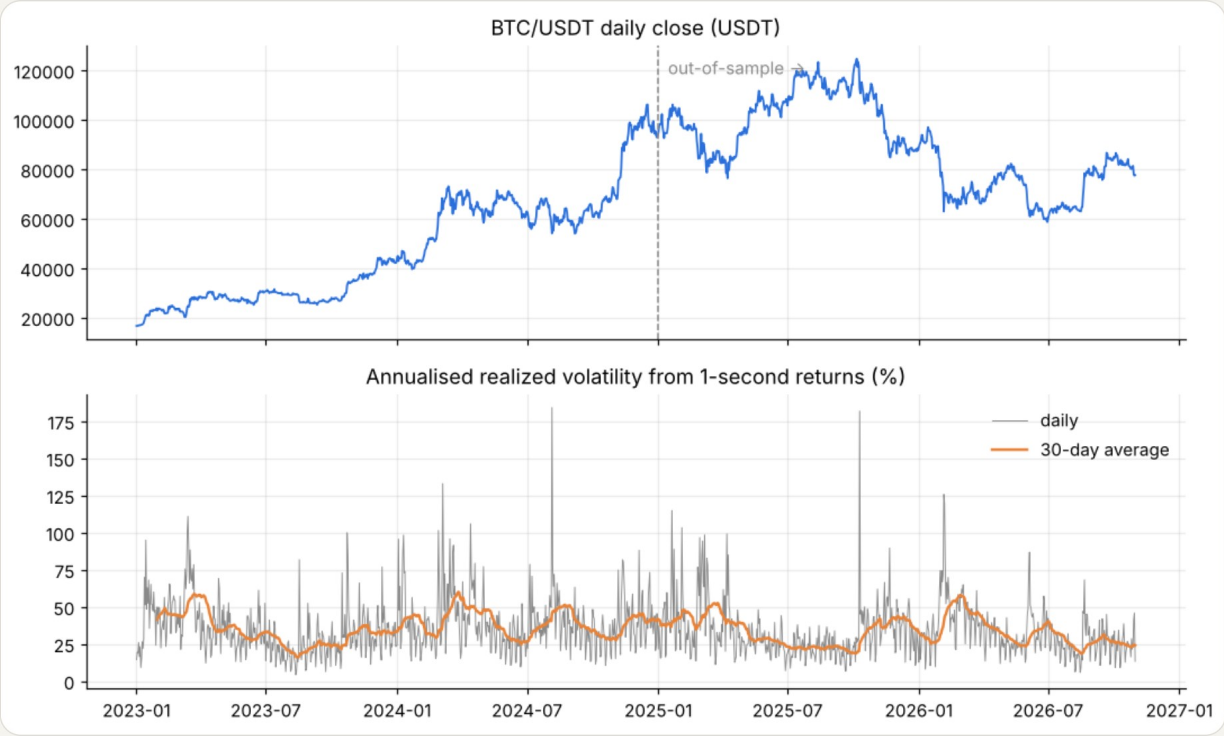
1. Download 46 monthly zips from Binance Public Data (bulk archive of the /api/v3/klines endpoint)
2. Verify each file against its published SHA-256 checksum
3. Standardise timestamps (ms until 2024, µs from Jan 2025), remove duplicates, flag rows failing OHLC logic
4. Fill missing seconds: carry the last price, zero volume
5. Aggregate to 1-minute bars, keeping 1-second features: realized variance, last-10-second return, active seconds

Data quality

1-second klines	120,951,229
Monthly files	46
Checksums passed	46 / 46
Duplicate rows	0
Rows failing OHLC checks	0
Missing seconds filled	8,771 (all Jan to Mar 2023)
1-minute bars produced	2,016,000

Source: data.binance.vision, BTCUSDT spot 1s klines

Market context: a bull market, then a falling one

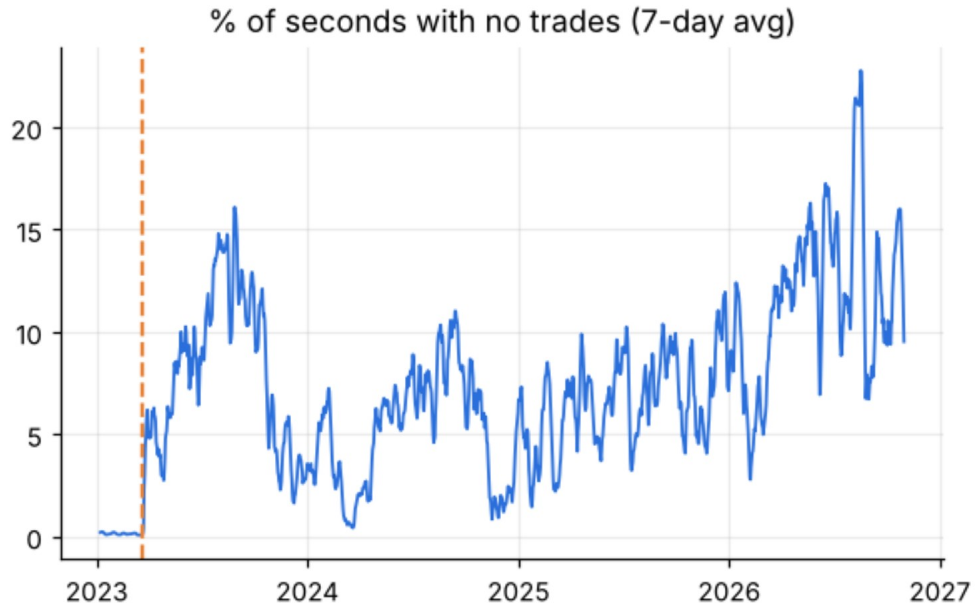


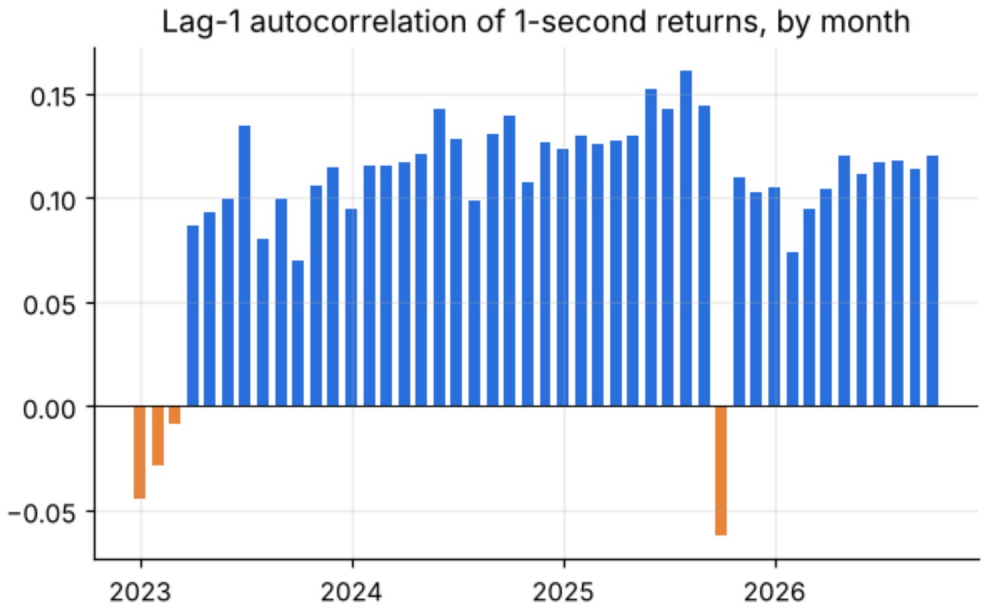
2023–24: BTC +155% and +112%. **2025–Oct 2026:** −7% and −13%. I design on the first period and test on the second, a harder market.

Date	Event	BTC that day
13 Mar 2023	SVB depositors backstopped, USDC re-pegs	+9%
10 Jan 2024	SEC approves spot bitcoin ETFs	+1%
5 Aug 2024	Yen carry-trade unwind	−7%
7 Apr 2025	US tariff sell-off	+1%
10 Oct 2025	China tariff threat, record liquidations	−8%
5 Feb 2026	Flash crash towards $60,000	−15%

Sources: The Block, CoinDesk, PYMNTS (links in notebook 02, section 7). Daily moves: own calculations

Pattern 2 · A structural break on 22 March 2023





Trades per day (3 weeks before vs after)

9.9m → 1.3m

USD volume per day

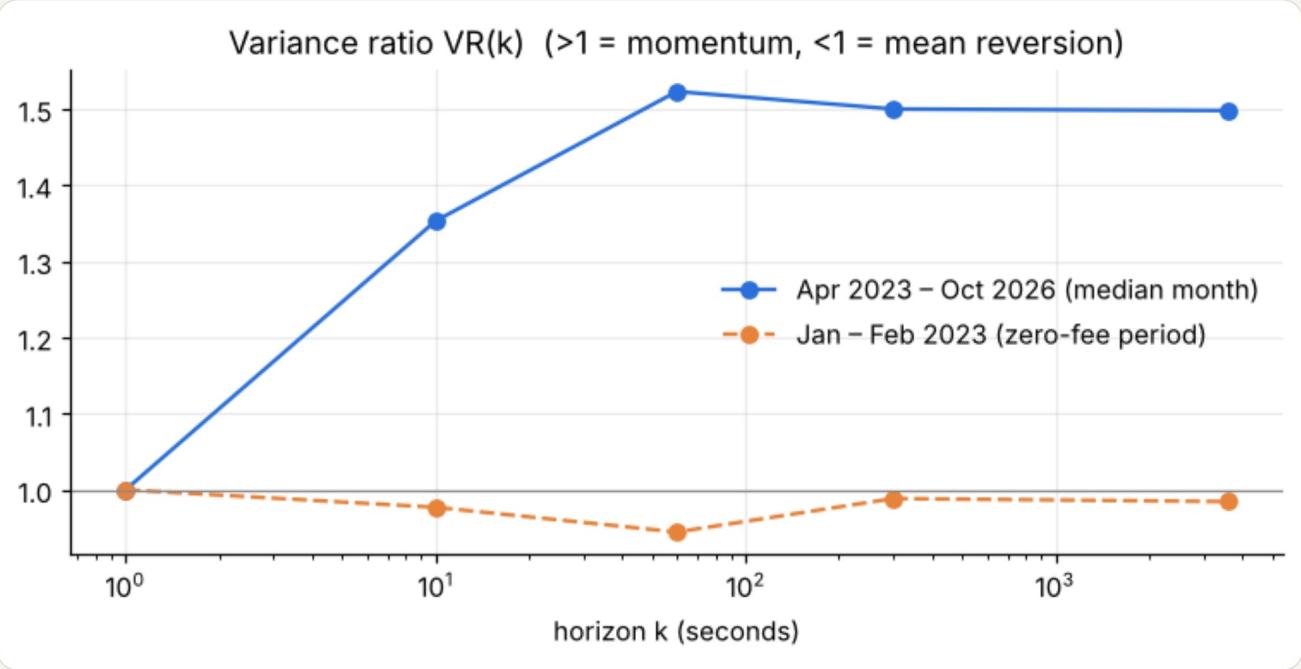
$9.7bn → $1.8bn

Seconds with no trades

0.1% → 5.6%

Event: Binance ended zero-fee BTC trading on 22 Mar 2023 (CoinDesk, 27 Mar 2023). Figures: own calculations

Pattern 3 · Short-term moves keep going

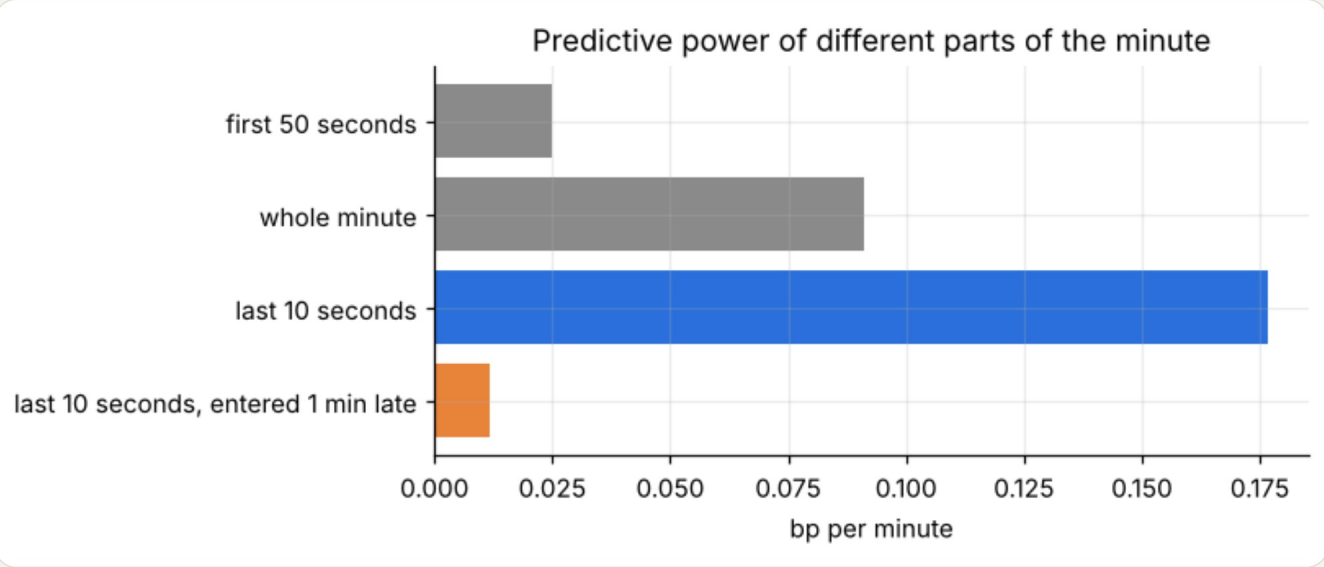


Variance ratio **VR(k)** = variance of k-second returns ÷ (k × variance of 1-second returns). A random walk gives 1.

Since April 2023 the median month has **VR ≈ 1.35** at 10 seconds and **≈ 1.5** from 1 to 60 minutes: moves continue.

In the zero-fee months it was slightly below 1. The Cowles–Jones ratio tells the same story: 0.94 before the break, 1.06 after. Momentum starts within seconds: VR(2 s) = 1.12, VR(5 s) = 1.26.

Where the signal sits: the last 10 seconds of each minute



Trading the direction of the **last 10 seconds** earns 0.18 bp per minute, twice the whole-minute signal and seven times the first 50 seconds.

Entering one minute later earns almost nothing: the information is used up fast.

This signal cannot be built from standard 1-minute klines.

The strategy

Last 10 seconds of the minute

Price up

→ long for the next minute

Last 10 seconds of the minute

Price down

→ short for the next minute

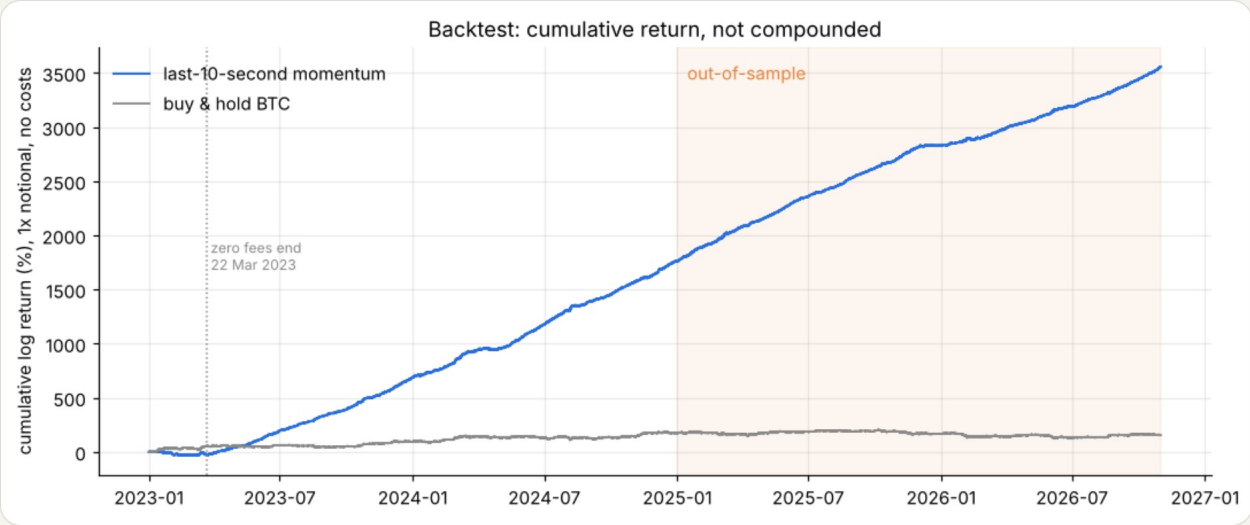
Last 10 seconds of the minute

Unchanged

→ flat

- One parameter, the 10-second window: selecting it on in-sample data from 2 to 30 seconds also gives 10 seconds, and every window is profitable
- A position set at the close of minute t earns the return from t to t+1, so no look-ahead
- In-sample 2023 to 2024, out-of-sample 2025 to Oct 2026; 1× notional, returns not compounded, costs excluded per the brief

Baseline results: profitable in and out of sample



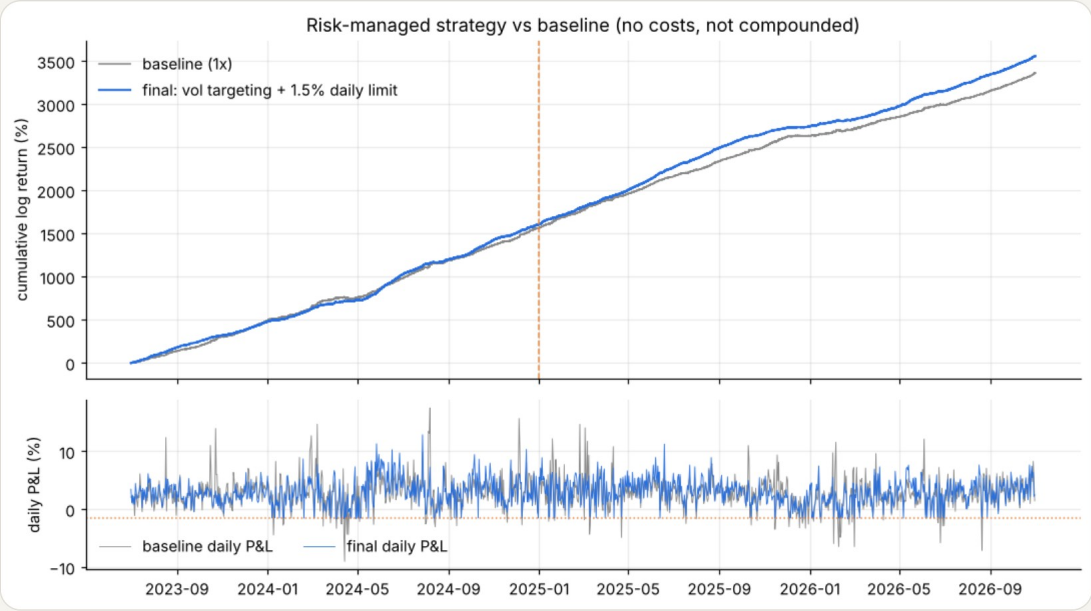
	Strategy 2023-24	Strategy 2025-26	Buy & hold 2025-26
Avg bp per minute	0.17	0.19	0.00
Breakeven cost, bp per unit	0.18	0.20	n/a
Return, simple %/yr	879	981	−10
Sharpe (annualised)	18.6	22.7	−0.2
Max drawdown, log %	−45*	−20	−78
Hit rate, non-zero minutes	51.0%	51.5%	50.1%

* Mostly Jan to Mar 2023, the zero-fee period; −20% from April 2023. Before transaction costs.

Robustness: it works where it should, and fails where it should

Check	Result	What it tells us
Every year	0.13 to 0.20 bp/min, all positive	Not driven by one lucky period
Zero-fee period (1 Jan to 21 Mar 2023)	−0.02 bp/min, loses money	Matches pattern 2: no momentum then
Long leg / short leg only (2025-26)	0.09 / 0.10 bp/min	Not just riding the bull market
Enter one minute late	0.02 / 0.00 bp/min	The edge decays within a minute
Trade only after large moves	Similar bp per trade, fewer trades	No gain from tuning, so kept simple
Signal window 2 to 30 s	All profitable; in-sample picks 10 s, best range 5–10 s	Result does not depend on the window
Data snooping (about 20 variants tried)	Bonferroni threshold t ≈ 3.0 vs t = 25 to 30	In-sample alone picks the same signal

Risk management: smaller bad days, same edge



1 · Volatility-targeted sizing. Size = σref / σrecent, capped at 2×. σrecent from the last hour of 1-second data; σref = median of the previous 90 days. Volatility clusters, and risk per minute grows faster than the edge in volatile markets.

2 · Daily loss limit. Stop for the day after −1.5%, about 0.65 × the strategy's daily standard deviation (2.3%). A 1% limit triggered on 1 day in 8; 1.5% triggers on about 1 day in 16.

2023–24* / 2025–26	bp per min	Worst day	Max drawdown	Sharpe
Baseline (1×)	0.20 / 0.19	−9.0% / −7.1%	−20% / −20%	22 / 23
+ volatility targeting (cap 2×)	0.21 / 0.21	−3.5% / −2.9%	−6.7% / −7.4%	34 / 35
+ 1.5% daily loss limit (final)	0.20 / 0.20	−1.7% / −1.7%	−6.6% / −7.6%	34 / 35

* Jul 2023 to Dec 2024. Sizing: Moreira and Muir (2017), volatility-managed portfolios. Before costs, not compounded

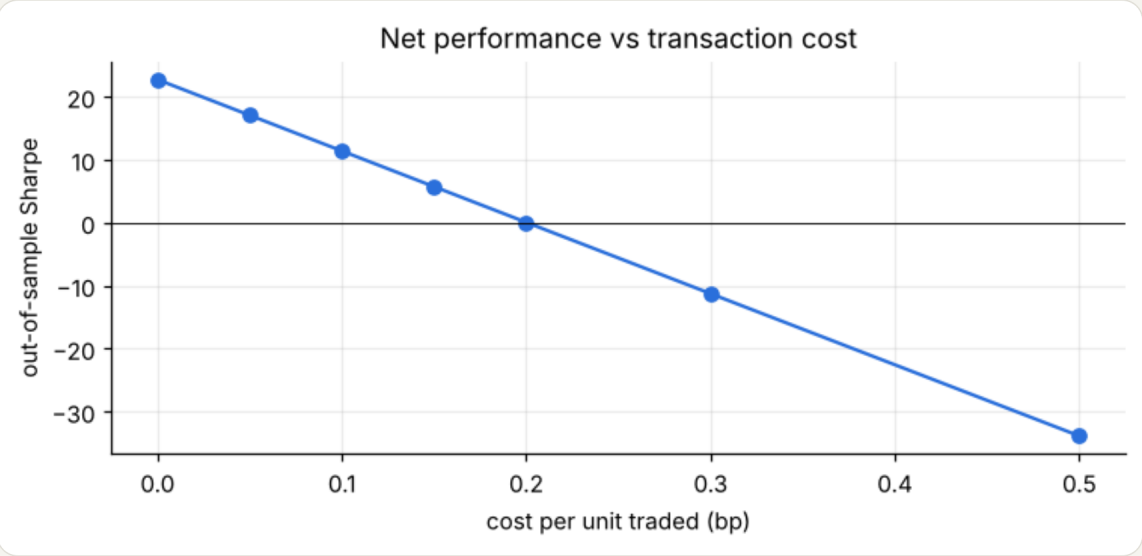
Comparison with classic BTC strategies

Strategy	Sharpe Jul 23–Dec 24	Sharpe 2025–26	Return 2025–26, %/yr	Max DD 2025–26, log %
Buy & hold	1.55	−0.24	−10%	−78
Time-series momentum, 30 days	0.68	0.37	+16%	−42
MA crossover, 20 / 50 days	0.45	0.68	+29%	−32
Donchian breakout, 20 days	1.02	0.05	+2%	−55
RSI(14) mean reversion	−1.25	0.09	+2%	−44
Hourly momentum	−1.87	−0.01	0%	−45
Last-10-second momentum, final	23.3	25.4	+1,066%	−7.6

Classic rules trade a few times a month and depend on the trend of each period. The last-10-second strategy is close to market-neutral and profitable in both periods, but it trades about 490,000 times a year, so this comparison only holds **before costs**.

Same data, daily rules applied from the next day. Moskowitz, Ooi and Pedersen (2012) for time-series momentum

Costs and practical use



About **490,000** position changes a year at about **0.2 bp** each. Net Sharpe reaches zero at a cost of **0.2 bp** per unit; Binance's standard spot fee is **10 bp** per side.

As a stand-alone strategy it suits only very low-cost, low-latency participants.

Practical use: execution timing. When a desk must buy or sell BTC anyway, for example to create or redeem ETP units, the last 10 seconds can decide whether to trade now or wait a minute. The fee is paid either way; the signal improves the fill.

Fee: Binance spot fee schedule, regular user 0.10% maker and taker

Conclusions and next steps

What I found

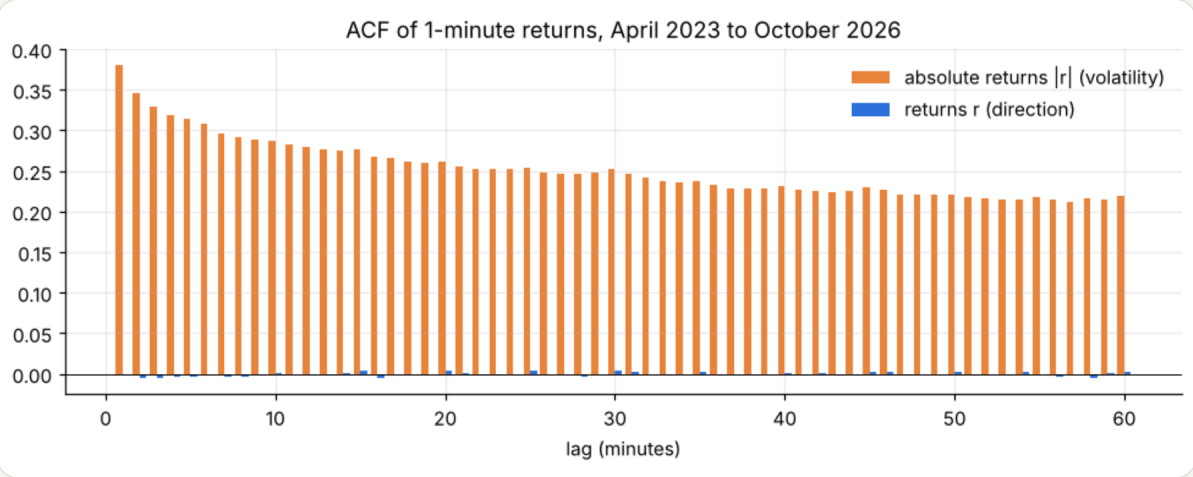
- A microstructure break on 22 Mar 2023, visible only in 1-second data
- Since then, short-term momentum concentrated in the last seconds of each minute
- A one-parameter signal, robust to the window (2 to 30 s), profitable before costs in every year, in and out of sample
- Risk management cuts the worst day to −1.7% with the same edge per trade
- Edge ≈ breakeven cost ≈ 0.2 bp: best used for execution timing

Next steps

- Volatility-scaled take-profit and stop-loss (ATR multiples) on each trade
- Latency test: entry 1, 2, 5 seconds after the minute close
- Order-book data: can passive (maker) orders capture the edge?
- Intraday volatility models: HAR-RV, deseasonalized intraday GARCH
- Formal data-snooping test: White's Reality Check / Hansen's SPA

Code and results: notebooks 01 (data), 02 (strategy) and 03 (signal window), in the GitHub repository

Appendix A · Formal tests: volatility clusters, the signal is significant



Ljung-Box, squared returns (60 lags)	$p \approx 0$
Engle ARCH LM test (daily, 10 lags)	$p = 6 \times 10^{-13}$
GARCH(1,1), daily: $\alpha + \beta$	0.99, half-life ≈ 70 days
Strategy mean, Newey-West t-stat	25 in-sample, 30 out-of-sample

Large moves follow large moves, so **volatility** is predictable. **Direction** from one minute close to the next is not (ACF ≈ 0), which is why the signal comes from inside the minute.

Appendix B · Random-walk tests: is the direction predictable?

Test	H0 and statistic	Result	Reading
Variance ratio Lo and MacKinlay (1988); CLM ch. 2	H0: random walk, VR(q) = 1 VR(q) = Var[r(q)] / (q·Var[r]) = 1 + 2 Σk=1q−1 (1 − k/q) ρ(k)	Median month after Apr 2023: VR(10 s) = 1.35, VR(60 s) = 1.52; zero-fee months slightly below 1	VR > 1 ⇔ positive autocorrelations: short-term momentum
Cowles–Jones ratio Cowles and Jones (1937)	H0: sequences and reversals equally likely, CJ = 1 CJ = Ns / Nr; z = (πs − ½) / √(¼ / N)	Zero-fee: 0.94 (z = −10.2) Apr 2023–2024: 1.057 (z = 23.5) 2025–Oct 2026: 1.062 (z = 25.9)	Sign of the last 10 s repeats in the next minute more often than it reverses
Ljung-Box Q on returns Box and Pierce (1970); Ljung and Box (1978)	H0: ρ(1) = … = ρ(m) = 0 Q(m) = T(T+2) Σk=1m ρ̂(k)² / (T − k) ~ χ²m	1-minute returns: Q(60) = 743, p ≈ 10^{-118}; but ρ̂(1) = −0.002	Significant only because T = 1.9 m; minute closes carry almost no linear predictability

Period: Apr 2023 to Oct 2026 unless stated. CLM = Campbell, Lo and MacKinlay (1997)

Appendix C · Volatility models and significance of the signal

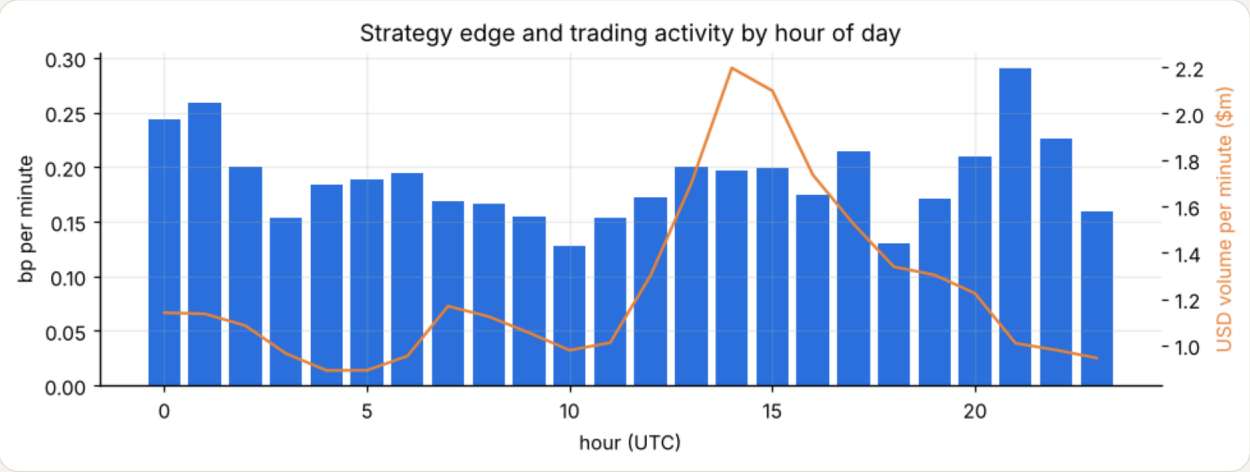
Test	H0 and statistic	Result	Reading
Ljung-Box Q on squared returns	H0: no autocorrelation in r^2 Q(m) as on the previous slide	1-minute: Q(60) = 553,834, p ≈ 0 ACF of $\|r\|$: 0.38 at lag 1, 0.22 at lag 60	Strong volatility clustering
ARCH LM test Engle (1982)	H0: no ARCH effects regress $\varepsilon^2 t$ on q own lags; LM = $T \cdot R^2 \sim \chi^2 q$	Daily, q = 10: LM = 79.7, p = 6×10^{-13}	Conditional variance is time-varying
GARCH(1,1), Student-t Bollerslev (1986)	$\sigma^2 t = \omega + \alpha\, \varepsilon^2 t-1 + \beta\, \sigma^2 t-1$ half-life = ln 0.5 / ln(α + β)	Daily: α = 0.058, β = 0.932, α + β = 0.990, half-life ≈ 70 days; t d.o.f. = 3.2	Very persistent volatility, fat tails
GJR-GARCH(1,1) Glosten, Jagannathan and Runkle (1993)	adds $\gamma\, \varepsilon^2 t-1\, 1\{\varepsilon t-1 < 0\}$; H0: γ = 0 (no leverage effect)	γ = 0.060, t = 1.59, p = 0.11 AIC 5727.5 vs 5730.8; BIC 5758.6 vs 5756.7	No significant leverage effect; BIC prefers plain GARCH
Newey-West t-test of strategy mean	H0: mean strategy return = 0, HAC standard errors (60 lags)	0.167 bp/min, t = 25.4 (in-sample) 0.187 bp/min, t = 29.9 (out-of-sample)	Edge is statistically very strong
GARCH residual checks	Ljung-Box (10 lags) on $z = \varepsilon/\sigma$ and z^2; ARCH LM on z	z: p = 0.45 z^2: p = 0.02; ARCH LM: p = 0.02	Most clustering captured; small remainder (jumps)

Daily models: Apr 2023 to Oct 2026, returns in %. Intraday GARCH hits α + β ≈ 1 (intraday seasonality), so daily data is used

Appendix D · Event days and time of day

Event day	BTC	Baseline	Final
13 Mar 2023	+9.2%	+2.2%	n/a
10 Jan 2024	+1.2%	0.0%	−0.3%
5 Aug 2024	−7.4%	+13.5%	+2.1%
7 Apr 2025	+0.9%	−0.3%	+2.1%
10 Oct 2025	−7.6%	+11.0%	+5.0%
5 Feb 2026	−15.1%	+11.6%	+2.6%

On the 15 most volatile days, the strategy was positive on 80% of them. Final sizing starts Jul 2023.



The edge is positive in every hour (UTC), between about 0.13 and 0.29 bp per minute, with no session where it disappears. I did not add a time-of-day filter.